

LLANO, TX, USA

WMO: 720110

Lat: 30.784N

Lon: 98.662W

Elev: 336

StdP: 97.35

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53983

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.7	-2.1	-12.3	1.4	3.8	-10.0	1.7	5.4	10.4	12.7	8.9	12.8	1.2	0	0.403

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.8	39.0	22.3	37.8	22.7	37.3	22.8	26.4	28.4	25.8	29.7	25.1	30.6	3.6	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.2	22.5	27.1	24.9	20.9	26.7	23.9	19.5	26.3	84.5	28.5	81.6	31.1	78.4	30.0	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.4	7.4	6.4	DB	-8.2	41.7	2.5	1.1	-10.0	42.4	-11.4	43.1	-12.9	43.7	-14.7	44.4	(4)
			WB	-9.1	26.6	2.4	0.7	-10.9	27.1	-12.3	27.5	-13.6	27.9	-15.3	28.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.0	9.4	12.0	16.0	19.9	23.9	28.0	29.3	29.8	26.3	20.7	14.5	10.2
	DBStd	8.29	4.93	5.50	4.89	4.27	3.57	2.16	1.97	2.03	2.93	4.60	5.17	4.86
	HDD10.0	193	69	38	8	1	0	0	0	0	0	1	17	59
	HDD18.3	1026	278	184	104	33	4	0	0	0	0	31	134	257
	CDD10.0	3860	52	94	195	297	430	541	600	614	488	332	153	64
	CDD18.3	1652	2	7	33	79	176	291	342	356	238	103	21	3
	CDH23.3	20174	55	128	371	858	1817	3593	4492	4842	2656	1078	224	59
CDH26.7	10599	9	29	101	318	804	1944	2621	2906	1379	426	52	9	
Wind	WSAvg	2.3	2.2	2.5	2.9	2.9	2.7	2.6	2.2	2.0	1.8	2.0	2.1	2.1
Precipitation	PrecAvg	695	33	47	46	58	90	81	47	55	65	73	54	43
	PrecMax	1056	149	154	177	278	191	225	151	284	173	185	228	290
	PrecMin	411	0	0	2	4	12	0	0	0	0	0	0	0
	PrecStd	142	31	37	35	49	53	57	41	55	45	47	45	46
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.8	30.1	32.1	36.0	37.3	39.1	40.9	41.2	38.8	34.1	31.2	27.7
		MCWB	14.9	17.9	16.3	19.4	22.6	22.7	21.9	22.6	21.3	20.8	20.7	16.3
	2%	DB	24.8	27.1	29.0	32.4	34.8	37.6	38.7	39.1	37.1	32.5	27.8	25.0
		MCWB	14.9	16.2	17.5	19.6	22.2	23.0	22.1	22.5	22.1	21.5	18.4	15.8
	5%	DB	22.2	24.1	27.5	30.2	32.7	36.3	37.6	38.0	35.1	31.2	26.2	22.4
		MCWB	14.5	15.8	17.3	19.6	22.2	22.9	22.7	22.8	22.5	20.9	18.0	15.7
	10%	DB	19.0	22.3	25.8	27.9	31.3	34.2	36.4	37.4	33.0	28.9	23.7	20.0
		MCWB	13.0	15.2	16.6	19.0	22.1	22.9	23.1	22.9	22.2	20.0	17.3	13.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.8	20.6	22.5	24.2	27.0	26.8	27.0	27.8	27.0	25.3	22.5	22.0
		MCDB	22.0	25.0	24.1	27.8	28.7	29.3	28.1	28.6	28.2	29.7	26.4	23.1
	2%	WB	18.0	19.3	20.5	22.9	25.1	25.8	25.9	26.3	25.8	24.0	21.2	19.0
		MCDB	21.0	23.7	24.7	27.0	30.0	30.6	29.2	28.0	29.4	28.0	24.7	22.4
	5%	WB	16.1	18.0	19.5	22.0	24.1	25.0	25.1	25.5	24.7	23.2	20.0	17.2
		MCDB	20.2	21.4	24.4	27.0	28.8	31.2	30.7	30.2	29.9	27.1	23.9	20.0
	10%	WB	13.6	16.5	18.5	21.1	23.4	24.4	24.6	24.8	23.9	22.5	18.9	15.1
		MCDB	19.0	20.6	23.0	25.9	28.6	30.7	30.6	30.8	29.3	26.4	22.9	19.0
Mean Daily Temperature Range	5% DB	MDBR	14.6	14.5	14.6	14.1	12.5	12.8	13.1	13.8	13.1	14.3	14.9	14.6
		MCDBR	19.6	18.7	16.6	17.0	15.1	14.9	15.1	15.8	15.6	15.4	16.8	19.1
		MCWBR	11.2	10.5	7.7	7.5	5.7	3.7	3.4	3.6	4.5	6.5	9.0	11.2
	5% WB	MCDBR	14.8	14.4	12.5	12.1	11.7	11.3	11.7	12.4	11.4	11.1	12.7	14.4
		MCWBR	10.4	9.5	7.1	6.6	5.4	3.9	3.7	4.2	4.4	5.8	8.3	10.5
Clear-Sky Solar Irradiance	taub	0.313	0.319	0.347	0.386	0.421	0.417	0.429	0.419	0.393	0.366	0.336	0.316	
	taud	2.518	2.508	2.441	2.354	2.311	2.369	2.377	2.385	2.406	2.468	2.485	2.515	
	Ebn at Noon	921	947	937	906	872	870	858	865	878	882	885	898	
	Edh at Noon	87	96	110	125	131	123	122	120	113	99	89	83	
All-Sky Solar Radiation	RadAvg	3.14	3.75	4.74	5.68	5.94	6.80	6.78	6.35	5.24	4.27	3.28	2.79	
	RadStd	0.34	0.56	0.43	0.52	0.62	0.51	0.55	0.37	0.48	0.63	0.42	0.37	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+77

Nomenclature: See separate page